

# DigiSign Identify PoC

## User Manual

A clear guide to preparing DigiSign, running the application, obtaining a bearer token, starting identity verification, and reviewing the trusted provider result.

|                          |                             |                             |                           |
|--------------------------|-----------------------------|-----------------------------|---------------------------|
| <b>1</b><br>Authenticate | <b>2</b><br>Choose scenario | <b>3</b><br>Verify identity | <b>4</b><br>Review result |
|--------------------------|-----------------------------|-----------------------------|---------------------------|

| Audience                                                           | Version         |
|--------------------------------------------------------------------|-----------------|
| DigiSign administrators, developers, testers, and first-time users | 1.1 - July 2026 |

### Proof of Concept

This application is intentionally small. Use staging credentials and authorised test participants only.

# Contents

Use the quick-start checklist for the shortest path, or go directly to the section you need.

| Section | Topic                                       | Page |
|---------|---------------------------------------------|------|
| 1       | About this manual                           | 3    |
| 2       | Quick-start checklist                       | 4    |
| 3       | Obtain DigiSign staging access              | 5    |
| 4       | Configure DigiSign Identify                 | 6    |
| 5       | Supported scenarios and checks              | 7    |
| 6       | Create DigiSign API credentials             | 8    |
| 7       | Install and build the PoC                   | 9    |
| 8       | Start the application and open the UI       | 10   |
| 9       | Authenticate in the UI                      | 11   |
| 10      | Choose a scenario and start verification    | 12   |
| 11      | Complete verification and review the result | 13   |
| 12      | Manual acceptance test                      | 14   |
| 13      | Troubleshooting                             | 15   |
| 14      | Optional phone test                         | 16   |
| 15      | Security and production limitations         | 17   |
| 16      | Glossary and references                     | 18   |

## Recommended reading order

Administrators: sections 3-6. PoC operators: sections 7-13. Test participants: sections 10-11.

# 1. About this manual

This manual explains how to run the DigiSign Identify Proof of Concept from start to finish. No previous DigiSign or ASP.NET knowledge is required.

## What this version proves

- The PoC can request a bearer JWT from DigiSign by using an accessKey and secretKey.
- An existing bearer JWT can be used instead of API keys.
- The UI can load available Identify scenarios from DigiSign.
- The PoC creates an identification and starts it.
- DigiSign opens outside an iframe in a new browser tab.
- After the browser returns, the PoC retrieves the authoritative identification status from the authenticated API.

## The complete journey

| 1. Prepare                           | 2. Start                                         | 3. Verify                                            | 4. Decide                                                       |
|--------------------------------------|--------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------|
| Create staging scenario and API key. | Authenticate, choose scenario, create and start. | User completes document, selfie, and liveness steps. | PoC displays DigiSign API status; refresh if review is pending. |

## Who does what?

| Role                   | Responsibility                                                                     |
|------------------------|------------------------------------------------------------------------------------|
| DigiSign administrator | Enables Identify, creates scenarios and API credentials, and reviews manual cases. |
| PoC operator           | Runs the app, enters settings, starts the test, and records the outcome.           |
| Verified person        | Follows DigiSign document, camera, selfie, and liveness instructions.              |

### Personal data

Do not send access keys, secret keys, JWTs, document photos, selfies, or extracted identity data by email or chat.

## 2. Quick-start checklist

- 1 Request a DigiSign staging account with Identify and API access.
- 2 Sign in to DigiSign staging administration.
- 3 Create or activate an Identify scenario. For an ID-card test, enable the relevant country and ID-card document type.
- 4 Create an API key and securely save its **accessKey** and **secretKey**.
- 5 Install the .NET 10 SDK, update the repository, and build the PoC.
- 6 Run the app and open **https://localhost:7025**.
- 7 Enter the API keys and select **Get bearer token**, or paste an existing JWT.
- 8 Select **Load available scenarios**, choose the scenario ID, and review the callback URL.
- 9 Select **Create and start verification**.
- 10 Complete DigiSign in the new tab, return to the PoC, and review the authoritative status.

### You will need

| Item                         | Where it comes from                                         |
|------------------------------|-------------------------------------------------------------|
| .NET 10 SDK                  | dotnet.microsoft.com                                        |
| DigiSign staging account     | Requested from DigiSign support                             |
| Identify scenario ID         | DigiSign staging administration or Load available scenarios |
| accessKey + secretKey        | DigiSign staging API keys                                   |
| Bearer JWT                   | Optional: supplied by you or requested by the PoC UI        |
| Identity document and camera | The authorised test participant                             |

#### Recommended first test

Use a desktop browser first. A phone cannot return to a callback hosted on your computer's localhost address.

## 3. Obtain DigiSign staging access

Testing and production are separate DigiSign environments. Start in staging. Staging has separate users, scenarios, API keys, URLs, and data.

### 3.1 Request the account

To: podpora@digisign.cz  
Subject: DigiSign Identify staging access for PoC

Please create a DigiSign staging account for a standalone DigiSign Identify PoC. We need Identify, REST API access, permission to create API keys, and scenario administration.  
Project: ECUPK Loaning System  
Purpose: standalone identity verification through /api/identifications

### 3.2 Confirm permissions

- 1 Open <https://app.staging.digisign.org/selfcare/login>.
- 2 Sign in with the account provided by DigiSign.
- 3 Confirm that you can see DigiSign Identify settings, verification scenarios, and API keys.
- 4 If a menu is missing, ask the workspace administrator or DigiSign support to enable the module or permission.

### Environment addresses

| Purpose        | Staging                  | Production       |
|----------------|--------------------------|------------------|
| Administration | app.staging.digisign.org | app.digisign.org |
| API            | api.staging.digisign.org | api.digisign.org |

#### Never mix environments

A staging key, scenario ID, or bearer token will not work against the production API, and vice versa.

## 4. Configure DigiSign Identify

Scenario configuration is a manual DigiSign administration task. The PoC sends only the selected scenario ID; the scenario controls which evidence and checks the user must complete.

### 4.1 Enable Identify

- 1 Open **Settings / DigiSign IDENTIFY / Settings**.
- 2 Enable the DigiSign Identify module.
- 3 Keep staging retention defaults unless the test plan defines different rules.

### 4.2 Create the scenario

- 1 Open **Settings / DigiSign IDENTIFY / Verification scenarios**.
- 2 Select **New scenario**.
- 3 Use a clear name, for example **ECUPK PoC - automatic Czech ID card**.
- 4 Set the scenario to Active.
- 5 Choose Automatic for document authenticity, selfie, face match, and liveness testing.
- 6 Enable the required countries and document types. For the user's ID-card scenario, enable **ID card**.
- 7 Review consent, evaluation, and retention settings, then save.

#### DigiSign staging administration - scenario example

|                          |                                     |                      |
|--------------------------|-------------------------------------|----------------------|
| <b>Name</b>              | ECUPK PoC - automatic Czech ID card |                      |
| <b>Method</b>            | Automatic                           | Recommended PoC mode |
| <b>Country</b>           | Czech Republic                      |                      |
| <b>Primary document</b>  | ID card                             |                      |
| <b>Selfie / liveness</b> | Included by Automatic mode          |                      |
| <b>State</b>             | Active                              |                      |

#### Scenario ID

The scenario UUID can be copied from DigiSign administration or loaded directly by the PoC UI after authentication.

## 5. Supported scenarios and checks

The selected DigiSign scenario determines the actual user-verification workflow. The PoC does not hard-code document checks.

| Method          | What happens                                                                            | PoC use                                               |
|-----------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------|
| No verification | Collects configured evidence without proving authenticity or the person.                | Not suitable for the main identity-verification test. |
| Manual          | Collects evidence; an administrator approves or denies the identification.              | Useful for testing an operational review queue.       |
| Automatic       | Runs configured automated document and person checks; uncertain cases may go to review. | Recommended for the main PoC.                         |

### ID-card scenario

#### Yes - supported

The PoC accepts any valid Identify scenario ID returned by DigiSign. An ID-card scenario works when the scenario is active, the account has access, and the document country/type is enabled in DigiSign.

### Checks commonly controlled by the scenario

| Check or evidence         | Behaviour                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------|
| Primary identity document | ID card, passport, driving licence, or other provider-supported types selected by the administrator. |
| Selfie                    | Optional in some collection/manual flows; included in Automatic identity verification.               |
| Liveness and face match   | Performed when enabled by the selected Automatic scenario/provider flow.                             |
| Allowed countries         | Restricts which issuing countries/documents the participant can select.                              |
| Manual review             | A completed result can remain for _review until an administrator decides.                            |
| Retention                 | Controls how long incomplete and completed records/evidence remain available.                        |

**Scenario versions:** existing identifications keep the scenario version used at creation; later changes affect new identifications.

## 6. Create DigiSign API credentials

- 1 In staging administration, open **Settings / For developers / API keys**.
- 2 Select **New API key**.
- 3 Grant access to authentication, scenarios, and identification endpoints needed by the PoC.
- 4 Create the key and immediately save both displayed values in a secure password manager.

| DigiSign staging administration - API key example |                                      |  |
|---------------------------------------------------|--------------------------------------|--|
| Key name                                          | ECUPK DigiSign PoC                   |  |
| accessKey                                         | XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX |  |
| secretKey                                         | Displayed once - save securely       |  |
| Environment                                       | Staging                              |  |

### What each value means

| Value      | Purpose                            | Lifetime                             |
|------------|------------------------------------|--------------------------------------|
| accessKey  | Identifies the DigiSign API key.   | Longer-lived                         |
| secretKey  | Proves possession of the API key.  | Longer-lived; shown only at creation |
| Bearer JWT | Authorizes API calls from the PoC. | Short-lived                          |

#### New in version 1.1

You no longer need PowerShell to exchange the keys. Enter accessKey and secretKey in the PoC UI and select Get bearer token. You can still paste an existing bearer JWT when preferred.

#### Secret handling

The secretKey is submitted to DigiSign to obtain the JWT, then cleared from the page model. It is not saved in appsettings or session.



## 7. Install and build the PoC

### 7.1 Install required software

| Software | Requirement             | Check                    |
|----------|-------------------------|--------------------------|
| .NET SDK | Version 10.x            | dotnet --version         |
| Git      | Current Windows version | git --version            |
| Browser  | Edge or Chrome          | Open normally            |
| Camera   | Built-in or connected   | Test with Windows Camera |

Download the SDK from <https://dotnet.microsoft.com/download>. Install the SDK, not only the runtime.

### 7.2 Get or update the source

```
cd C:\Projects
git clone https://github.com/Ota85/PoC_DigiSign.git
cd C:\Projects\PoC_DigiSign\DigiSignPoC
```

If the repository already exists:

```
cd C:\Projects\PoC_DigiSign
git pull origin main
cd DigiSignPoC
```

### 7.3 Trust local HTTPS and build

```
dotnet dev-certs https --trust
dotnet build
```

#### Expected result

The build ends with Build succeeded and zero errors. The current PoC also builds with zero warnings.

## 8. Start the application and open the UI

```
cd C:\Projects\PoC_DigiSign\DigiSignPoC
dotnet run --launch-profile https
```

Keep the PowerShell window open. When it says **Now listening on: https://localhost:7025**, open that address in Edge or Chrome.

### The home page has four sections

| 1. DigiSign environment |                                                               |                                                      |
|-------------------------|---------------------------------------------------------------|------------------------------------------------------|
| Base URL                | <input type="text" value="https://api.staging.digisign.org"/> | <input type="checkbox"/> Use staging for PoC testing |

| 2. Authentication |                                                            |  |
|-------------------|------------------------------------------------------------|--|
| Bearer token      | <input type="text" value="Optional existing JWT"/>         |  |
| accessKey         | <input type="text" value="Optional when JWT is supplied"/> |  |
| secretKey         | <input type="text" value="Optional when JWT is supplied"/> |  |
|                   | <input type="button" value="Get bearer token"/>            |  |

| 3. Identify scenario |                                                          |  |
|----------------------|----------------------------------------------------------|--|
| Scenario ID          | <input type="text" value="Enter or load from DigiSign"/> |  |
|                      | <input type="button" value="Load available scenarios"/>  |  |

| 4. Identification settings |                                                              |                                                  |
|----------------------------|--------------------------------------------------------------|--------------------------------------------------|
| Name                       | <input type="text" value="ECUPK DigiSign PoC test"/>         |                                                  |
| Link expiration            | <input type="text" value="0"/>                               | <input type="checkbox"/> 0 uses provider default |
| Redirect URL               | <input type="text" value="https://localhost:7025/Callback"/> |                                                  |
|                            | <input type="button" value="Create and start verification"/> |                                                  |

## 9. Authenticate in the UI

### Option A - request a JWT with API keys

- 1 Confirm the Base URL matches the environment where the API key was created.
- 2 Leave Bearer token empty.
- 3 Enter accessKey and secretKey.
- 4 Select **Get bearer token**.
- 5 Confirm the success message. The token field is filled and the expiry time is shown when supplied by DigiSign.

| Authentication - API-key path |                             |                         |
|-------------------------------|-----------------------------|-------------------------|
| <b>Bearer token</b>           | eyJ... filled after success | You can show or copy it |
| <b>accessKey</b>              | Your staging access key     |                         |
| <b>secretKey</b>              | Cleared after request       |                         |
|                               | <b>Get bearer token</b>     |                         |

### Option B - use an existing bearer JWT

- 1 Paste the complete JWT into Bearer token.
- 2 Leave accessKey and secretKey empty.
- 3 Continue to scenario selection or start verification.

#### Automatic fallback

If Bearer token is empty but both API-key fields are present, Load available scenarios and Create and start verification can obtain a token automatically.

#### Do not capture credentials

Avoid screenshots while accessKey, secretKey, or bearer JWT is visible. Never paste them into a bug report or work item.

# 10. Choose a scenario and start verification

## 10.1 Load or enter a scenario

- 1 Select **Load available scenarios**.
- 2 The PoC calls DigiSign and displays scenarios available to the authenticated account.
- 3 Copy/select the ID of the required active scenario, such as the configured ID-card scenario.
- 4 If scenario loading is unavailable, paste the scenario UUID manually.

## 10.2 Review identification settings

| Field           | What to enter                                                                               |
|-----------------|---------------------------------------------------------------------------------------------|
| Name            | A clear test label visible in DigiSign administration.                                      |
| Link expiration | 0 for the provider default (normally 5 minutes), or a supported positive number of minutes. |
| Redirect URL    | Absolute URL. For desktop local testing: https://localhost:7025/Callback                    |

## 10.3 Create and start

- 1 Allow pop-ups for localhost.
- 2 Select **Create and start verification**.
- 3 The PoC creates the identification, starts it, displays the provider ID and validity, and opens identifyUrl in a new tab.
- 4 If the browser blocks the tab, select the displayed fallback link.

### No iframe

DigiSign document verification is intentionally opened in a separate window or tab. The PoC does not embed it in an iframe.

## What the PoC stores temporarily

For the callback, the in-memory session keeps the selected API Base URL, bearer JWT, and identification ID for up to two hours. Restarting the app or losing the browser session can remove this correlation.

# 11. Complete verification and review the result

## 11.1 In the DigiSign tab

- 1 Review and accept the required consent.
- 2 Choose an allowed identity document, such as ID card.
- 3 Photograph the document with good lighting and all edges visible.
- 4 Complete selfie and liveness instructions when required by the scenario.
- 5 Submit and wait for DigiSign processing.
- 6 DigiSign returns the browser to the configured callback URL.

## 11.2 On the PoC callback page

| Verification Result     |                                       |              |
|-------------------------|---------------------------------------|--------------|
| Authoritative status    | approved                              | Example only |
| Identification ID       | Provider UUID                         |              |
| Approval mode           | automatic                             |              |
| Completed / approved at | Provider timestamps                   |              |
| Callback parameters     | Displayed for technical analysis only |              |
|                         | Refresh DigiSign status               |              |

The callback page calls **GET /api/identifications/{id}** with the stored JWT. The displayed provider status is therefore the trusted DigiSign API result, not a decision inferred from browser query parameters.

## Common result meanings

| Status              | Meaning                                                                                  |
|---------------------|------------------------------------------------------------------------------------------|
| approved            | DigiSign approved the identification.                                                    |
| completed           | The participant submitted evidence; this alone is not approval.                          |
| for_review          | An administrator must review and decide. Use Refresh DigiSign status after the decision. |
| denied              | The identification failed or was rejected.                                               |
| cancelled / expired | The flow was cancelled or not completed before expiry.                                   |

## 12. Manual acceptance test

Record Pass, Fail, or Not tested. Never copy credentials or personal evidence into the test record.

| Test           | Expected result                                                                         | Result |
|----------------|-----------------------------------------------------------------------------------------|--------|
| Authentication | Existing JWT works; API keys can optionally obtain a JWT in the UI.                     | _____  |
| AK-01          | Scenario ID, absolute callback URL, and display name are configurable.                  | _____  |
| AK-02          | Identification is created and provider ID is displayed.                                 | _____  |
| AK-03          | Start returns identifyUrl and optional validTo.                                         | _____  |
| AK-04          | DigiSign opens outside an iframe; fallback link works.                                  | _____  |
| AK-05          | Callback shows API status plus technical query parameters.                              | _____  |
| AK-06          | 0 uses default; supported positive LinkExpiration is accepted.                          | _____  |
| AK-07          | No real credential exists in Git, appsettings, logs, or screenshots.                    | _____  |
| AK-08          | Environment, scenario, flow, limitations, and questions are recorded.                   | _____  |
| End-to-end     | Participant completes provider flow and result becomes approved, denied, or for_review. | _____  |

### Suggested test record

Date/time:  
Tester and environment:  
Scenario name / ID / approval mode:  
Document types enabled:  
Authentication path: existing JWT / API keys  
Token request result (no token value):  
Scenario load result:  
Identification ID:  
New-tab and fallback behaviour:  
Document / selfie / liveness observed:  
Callback API status:  
Final administration status:  
Open production questions:

## 13. Troubleshooting

| Problem                   | Likely cause                                                               | What to do                                                              |
|---------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Token request returns 401 | Wrong key, wrong environment, expired/revoked key, or missing permission.  | Re-enter both keys and confirm staging/production Base URL.             |
| Bearer token rejected     | Expired JWT or environment mismatch.                                       | Obtain a fresh JWT from the matching environment.                       |
| No scenarios loaded       | Token lacks permission, Identify is disabled, or no scenario is available. | Check DigiSign administration and account permissions.                  |
| Scenario ID rejected      | Wrong UUID, inactive scenario, or ID from another environment.             | Load scenarios in the UI and use an active Identify scenario.           |
| Create returns 400        | Invalid callback, scenario, name, or expiration.                           | Use an absolute callback and LinkExpiration 0 for the first test.       |
| No new tab opens          | Browser blocked the pop-up.                                                | Allow pop-ups or use the fallback link.                                 |
| Camera unavailable        | Permission denied or camera used elsewhere.                                | Allow camera and close Teams, Zoom, or Camera.                          |
| Callback has no session   | App restarted, cookies lost, different browser/device, or session expired. | Return to the original browser and start a new verification.            |
| Status stays for_review   | Administrator decision is required.                                        | Review in DigiSign administration, then select Refresh DigiSign status. |
| Phone callback fails      | Phone localhost points to the phone.                                       | Use an approved public HTTPS callback.                                  |
| Certificate warning       | Local dev certificate is not trusted.                                      | Run dotnet dev-certs https --trust.                                     |

### Logs

The console shows safe diagnostic messages and HTTP status codes. Review logs before sharing because URLs, IDs, or personal data may still be sensitive.

## 14. Optional phone test

A phone often provides a better document/selfie experience, but it cannot return to **https://localhost:7025/Callback** on your computer. The callback must be publicly reachable over HTTPS.

### Security approval required

Use only an organisation-approved tunnel or temporary test deployment. Do not expose this PoC publicly without access controls and approval. Remove the endpoint immediately after testing.

### High-level process

- 1 Start the PoC on the development computer or approved test host.
- 2 Expose it through an approved temporary public HTTPS address.
- 3 Enter the public **https://your-test-host/Callback** in the Redirect URL field.
- 4 Open the public PoC start page using the same phone/browser session that will receive the callback.
- 5 Authenticate, create, and start the identification.
- 6 Complete DigiSign and confirm return to the public callback.
- 7 Verify that the callback can retrieve the authoritative provider status.
- 8 Stop the tunnel or remove the deployment immediately after the test.

### Why localhost fails

The word **localhost** always means this device. On the development computer it means the computer; on the phone it means the phone, where the PoC is not running.

### Session continuity

The callback depends on the PoC session cookie. Opening the verification on a different browser/device can lose the session correlation.



# 15. Security and production limitations

## What this PoC now does

- Obtains a short-lived JWT from API keys when requested.
- Clears secretKey after the token request.
- Stores bearer JWT and identification correlation only in in-memory session.
- Retrieves the authoritative identification object after callback.

## What production still needs

- Secure server-side credential storage, key rotation, and controlled token renewal.
- Persistent correlation between DigiSign identification and the authenticated application user.
- Explicit business rules for approved, for\_review, denied, cancelled, and expired.
- Reliable polling and/or authenticated provider notifications.
- Database persistence, audit trail, access control, retention, and deletion procedures.
- Secure public callback hosting, trusted-host validation, CSRF/session design, and multi-instance session storage.
- Personal-data impact assessment, approved consent, monitoring, retry, support, and incident procedures.
- Automated tests and provider-contract monitoring.

### **Do not expose the PoC as a production service**

The UI accepts API credentials and keeps the JWT in an in-memory session for convenience. This is acceptable for a controlled PoC, but production authentication and identity-result processing require a dedicated security design.

## Trusted result rule

The PoC correctly calls DigiSign's authenticated identification endpoint after return. Production must additionally enforce the application's own user correlation, required checks, scenario/version rules, and final business decision.

## 16. Glossary and references

| Term                   | Plain-language meaning                                                                                |
|------------------------|-------------------------------------------------------------------------------------------------------|
| PoC                    | Proof of Concept: a small application used to confirm the provider flow.                              |
| Identify scenario      | DigiSign rules defining documents, selfie, verification method, countries, evaluation, and retention. |
| Identification         | One concrete identity-verification attempt.                                                           |
| accessKey / secretKey  | Longer-lived API credentials used to request a short-lived JWT.                                       |
| JWT / bearer token     | Temporary credential sent in the Authorization header.                                                |
| identifyUrl            | DigiSign-hosted page where the participant performs verification.                                     |
| redirectUrl / callback | Address to which DigiSign returns the browser.                                                        |
| Authoritative status   | Status retrieved from DigiSign's authenticated API, not trusted from browser parameters.              |
| Liveness               | Check that the selfie comes from a live person rather than a photo or recording.                      |

### Useful references

- DigiSign staging administration: <https://app.staging.digisign.org>
- DigiSign staging API: <https://api.staging.digisign.org>
- DigiSign production administration: <https://app.digisign.org>
- DigiSign production API: <https://api.digisign.org>
- PoC repository: [https://github.com/Ota85/PoC\\_DigiSign](https://github.com/Ota85/PoC_DigiSign)

### Support

DigiSign staging, module, scenario, API-key, or permission questions: **podpora@digisign.cz**.

PoC source-code questions: use the ECUPK team's normal repository and work-item process.

#### End of manual

Document version 1.1 - updated for UI token retrieval, scenario discovery, and authoritative callback status.